

Chapter 6: A Tour of the Cell

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CURRICULUM MODULES INCLUDED

Module 0: Chapter 6 Content Introduction (Cell Theory & Cell Types) **Module 1:** Chapter 6 The Cell (Organelle Structure & Cellular Dynamics)

Module 0: Chapter 6 Content Introduction

Source: Chapter 6 Content Introduction_ BSC2010C-26Fall 0001.html

Chapter 6 Content Introduction

Learning Objectives

- Compare prokaryotic and eukaryotic cells
- Compare eukaryotic animal cells with eukaryotic plant cells
- Identify organelles
- Understand organelles functionality in maintaining cellular life

Chapter 6: Tour of the Cell

Cell Theory (remember the definition of theory?) states the following:

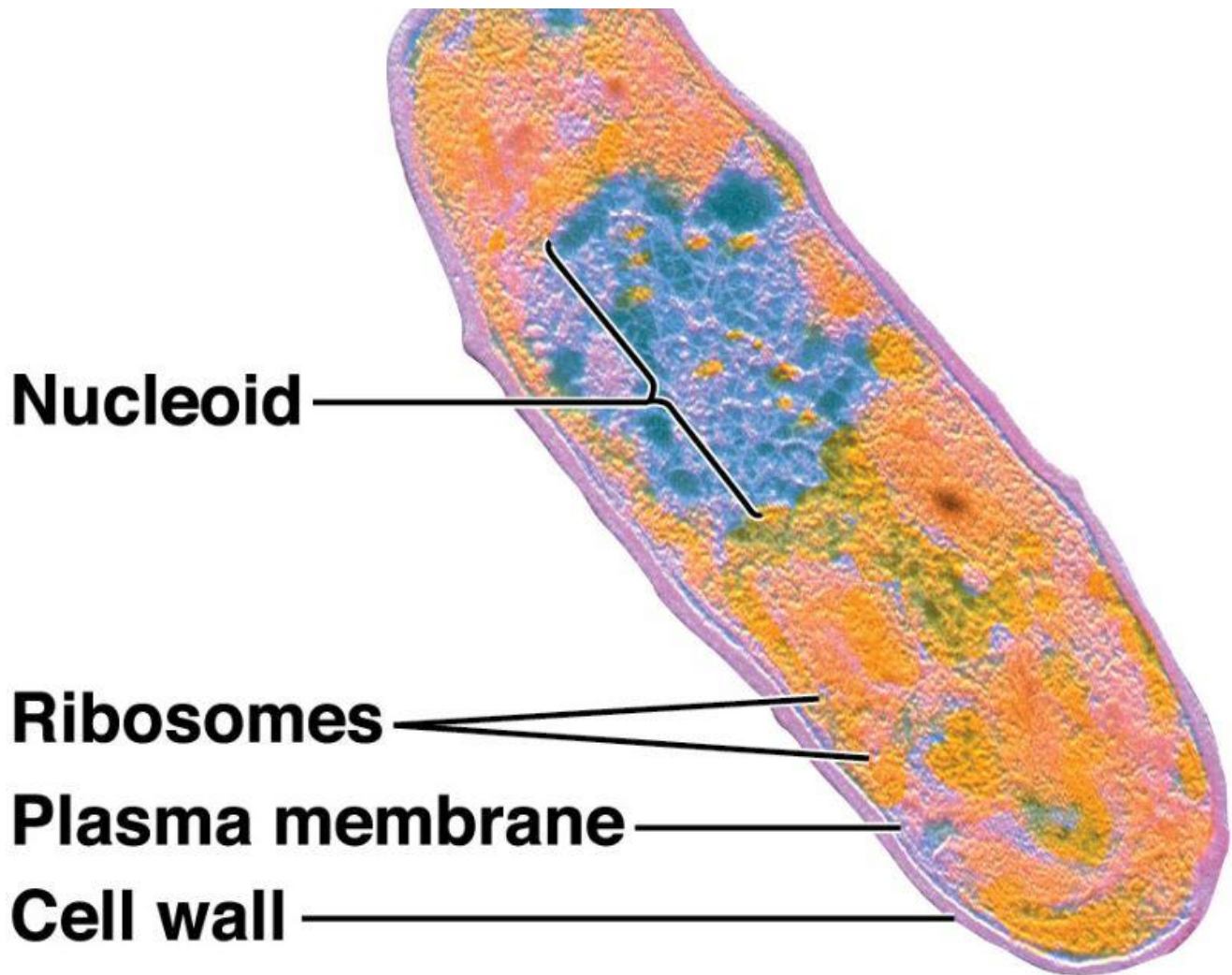
- all living things are composed of one or more cells
- cells are the smallest unit of life
- new cells come from existing cells through cell division

All cells can be divided into one of two types: Prokaryotes and Eukaryotes. Prokaryotes are "simple" single-celled bacteria from the Archaea and Bacteria domains. Eukaryotes can be single cell organisms or multicellular organisms from the Eukarya domain; they include protists, fungi, plants, and animals. All cells, both prokaryotes and eukaryotes have a few things in common: plasma membrane, DNA, ribosomes, and a cytosol. But prokaryotes and eukaryotes are very different from each other in many ways.

Prokaryotes are always single celled and they have a rather simple structure. They have a cell wall and plasma membrane that surround the cell; these are barriers between the cell and its external environment. The cytoplasm is the space inside the plasma membrane and it contains a watery fluid, called the cytosol, as well as several other structures that allow the organism to live and reproduce. Specifically, there are ribosomes, which synthesize

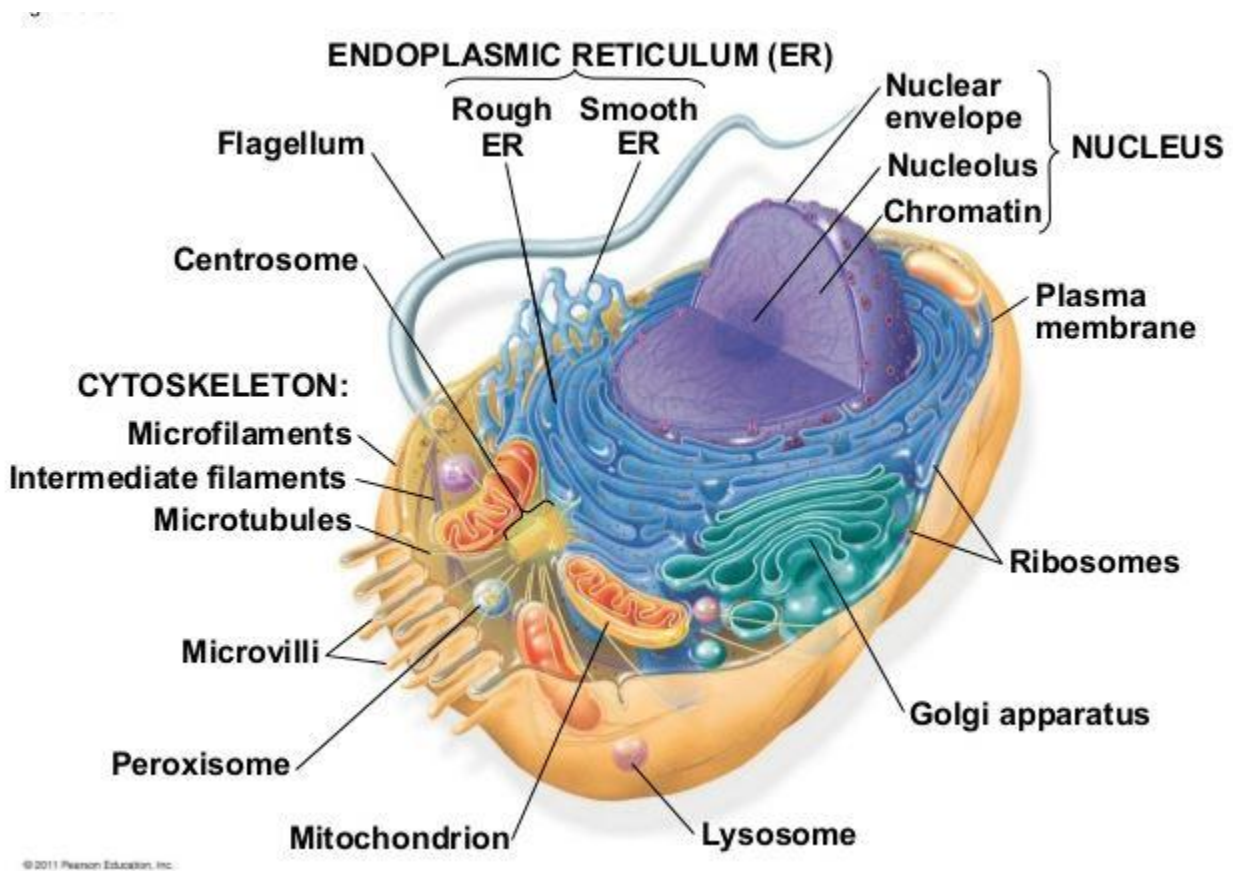
proteins, and a nucleoid which is the region where the cell's DNA can be found. Many prokaryotes have flagella or cilia to help them move about in their environment. Prokaryotes do not have any membrane-bound organelles.

Prokaryotic cell:



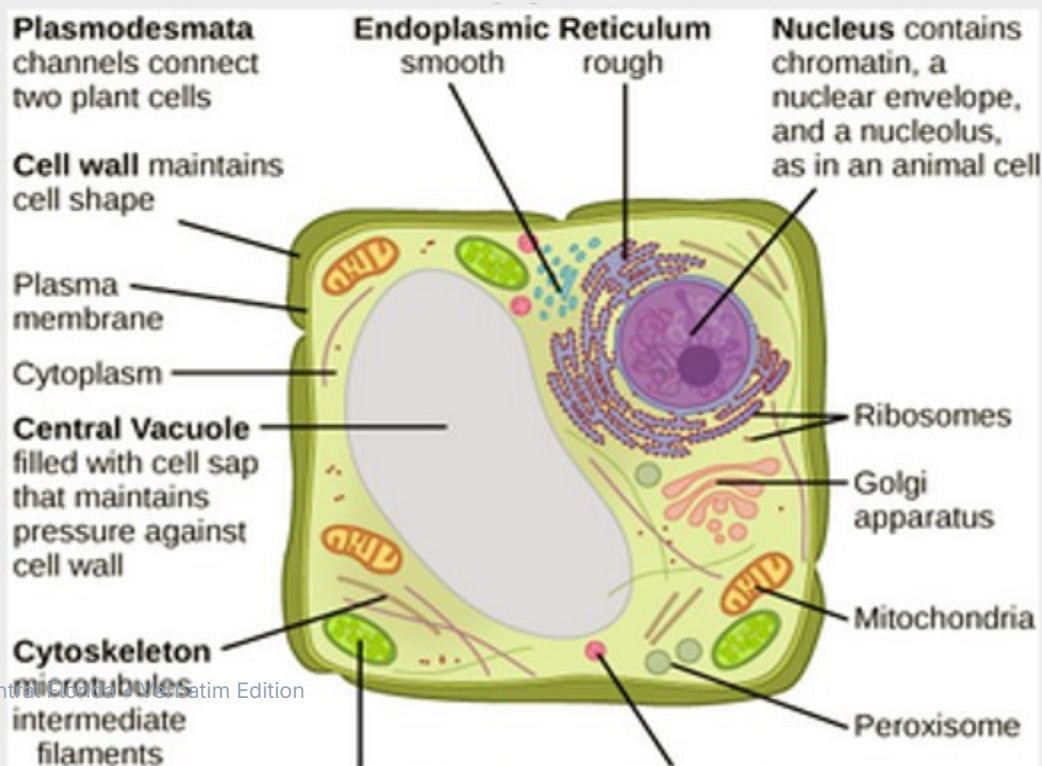
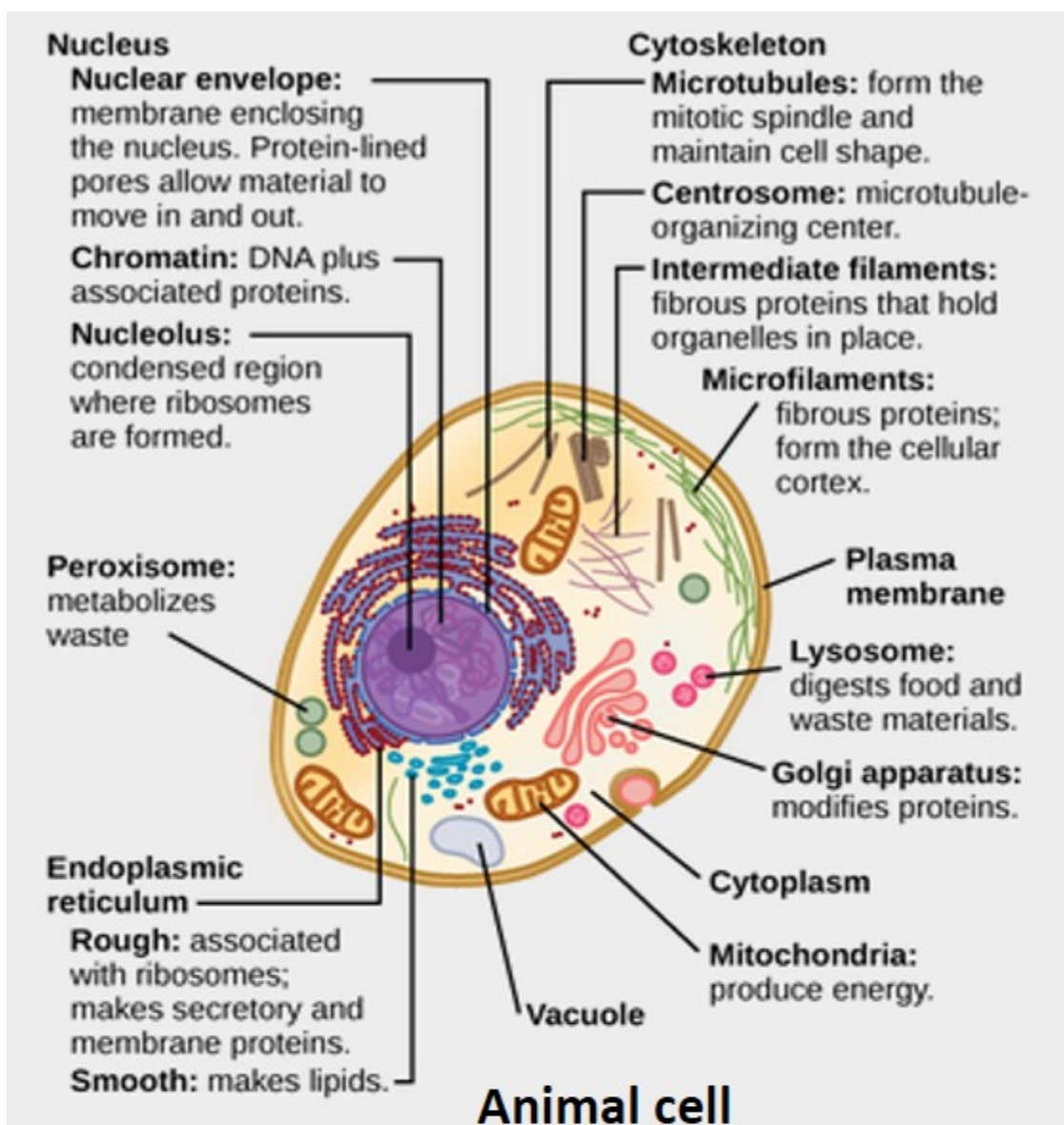
Eukaryotic cells are larger than prokaryotic cells and they have a more complex structure. All eukaryotic cells have a plasma membrane but only certain species of organisms have a cell wall. The cytoplasm is the space inside the plasma membrane and it contains a watery fluid, called the cytosol. Like prokaryotes, eukaryotic cells also contain ribosomes. But, unlike prokaryotes, eukaryotic cells are compartmentalized into membrane-bound organelles, each type of organelle has its own unique structure and function. Some examples of organelles are the **nucleus**, which houses the cells DNA, **mitochondria**, which produce ATP, the **Golgi apparatus**, which help the cell sort and ship proteins and lipids, and several other organelles what we will talk about in great detail in lecture.

Eukaryotic cell:



Cell Type Identification

Now that you have a basic understanding of the difference between prokaryotes and eukaryotes, you are ready to learn the differences between animal cells and plant cells. Although plant and animal cells have many organelles and cellular structures in common, there are a few organelles that are unique to plant cells only, and a few organelles that are unique to animal cells only. Plant cells have a cell wall outside their plasma membrane, chloroplasts, and a central vacuole; animal cells do not have these. Animal cells have lysosomes and centrosomes; plant cells do not have these. Take a look at the images below to see the organelles that plant and animal cells have in common. In lecture, we will learn about the structure and function of all cellular organelles and structures.



[Animal cell or Plant cell?](#)

[Cellular Organelles](#)

CHAPTER READING

[Chapter 6 Textbook Reading](#)

CHAPTER LINKS

Chapter 6 [PowerPoint](#) ([PDF](#)) [This powerpoint should follow very closely with the virtual lectures linked below]

Virtual Lecture: [Part 1](#) and [Part 2](#)

[Chapter 6 Post-Quiz](#)

[Chapter 6 Simple Practice Problems](#)

Module 1: Chapter 6 The Cell

Source: Chapter 6 The Cell_ BSC2010C-26Fall 0001.html

Chapter 6 The Cell

Learning Objectives

By the end of this section, you will be able to do the following:

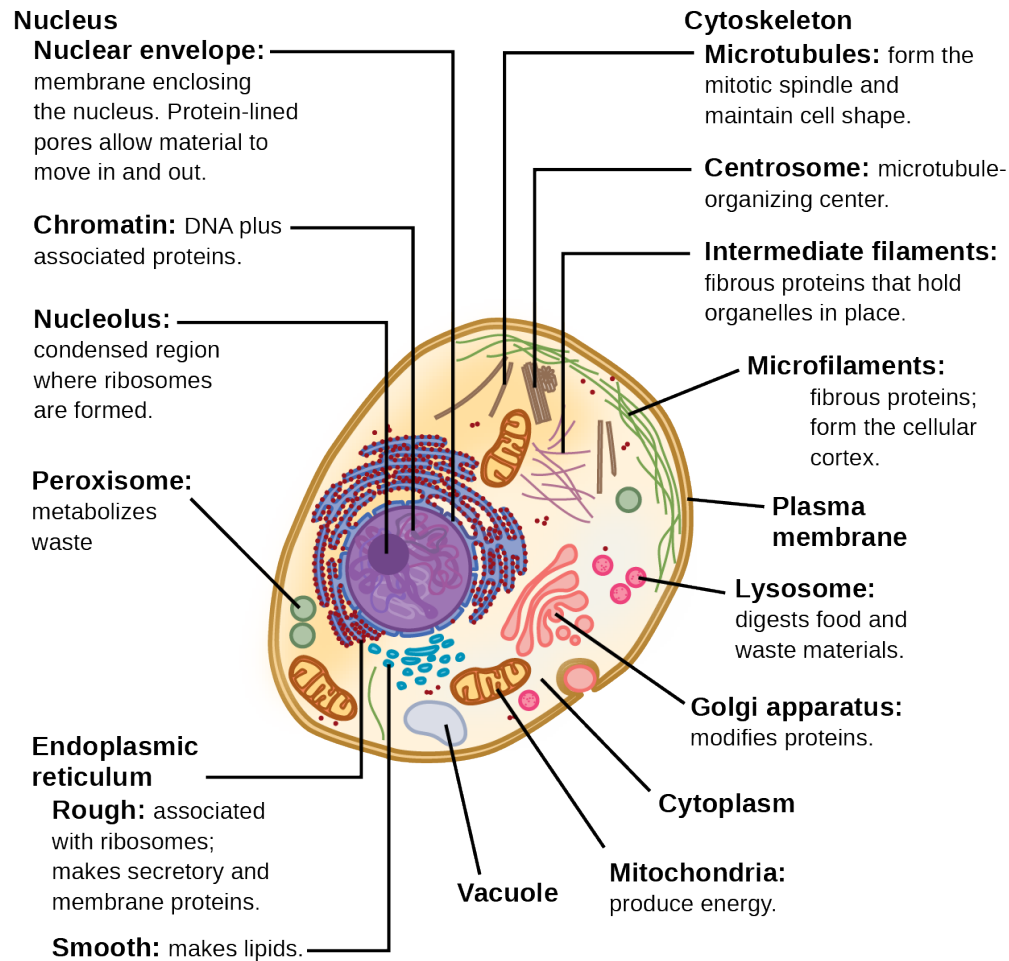
- Describe the structure of eukaryotic cells
- Compare animal cells with plant cells
- Summarize the functions of the major cell organelles

Eukaryotic cells have: 1) a membrane-bound nucleus; 2) numerous membrane-bound organelles such as the endoplasmic reticulum, Golgi apparatus, chloroplasts, mitochondria, and others; and 3) multiple linear chromosomes. Because a membrane surrounds eukaryotic cell's nucleus, it has a "true nucleus." The word "organelle" means "little organ," and, as we already mentioned, organelles have specialized cellular functions, just as your body's organs have specialized functions.

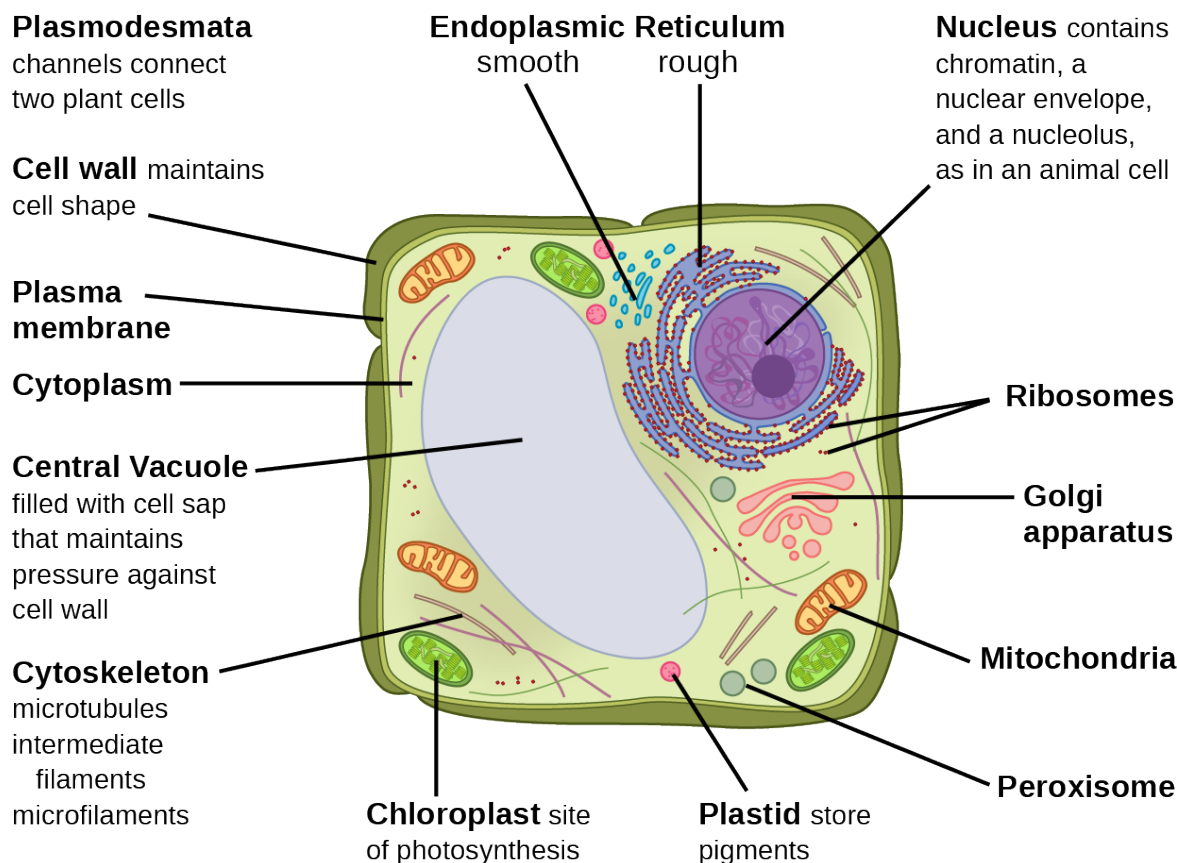
At this point, it should be clear to you that eukaryotic cells have a more complex structure than prokaryotic cells. Organelles allow different functions to be compartmentalized in different areas of the cell. Before turning to organelles, let's first examine two important components of the cell: the plasma membrane and the cytoplasm.

VISUAL CONNECTION

Visual Connection



Part a: This illustration shows a typical eukaryotic animal cell, which is egg shaped. The fluid inside the cell is called the cytoplasm, and the cell is surrounded by a cell membrane. The nucleus takes up about one-half...



Part b: This illustration depicts a typical eukaryotic plant cell. The nucleus of a plant cell contains chromatin and a nucleolus, the same as an animal cell. Other structures that the plant cell has in common with the a...

Part a: This illustration shows a typical eukaryotic animal cell, which is egg shaped. The fluid inside the cell is called the cytoplasm, and the cell is surrounded by a cell membrane. The nucleus takes up about one-half...

Figure 4.8 These figures show the major organelles and other cell components of (a) a typical animal cell and (b) a typical plant cell. The plant cell has a cell wall, chloroplasts, plastids, and a central vacuole—structures not in animal cells. Most cells do not have lysosomes or centrosomes.

The Plasma Membrane

Like prokaryotes, eukaryotic cells have a plasma membrane ([Figure 4.9](#)), a phospholipid bilayer with embedded proteins that separates the internal contents of the cell from its surrounding environment. A phospholipid is a lipid molecule with two fatty acid chains and a phosphate-containing group. The plasma membrane controls the passage of organic molecules, ions, water, and oxygen into and out of the cell. Wastes (such as carbon dioxide and ammonia) also leave the cell by passing through the plasma membrane.

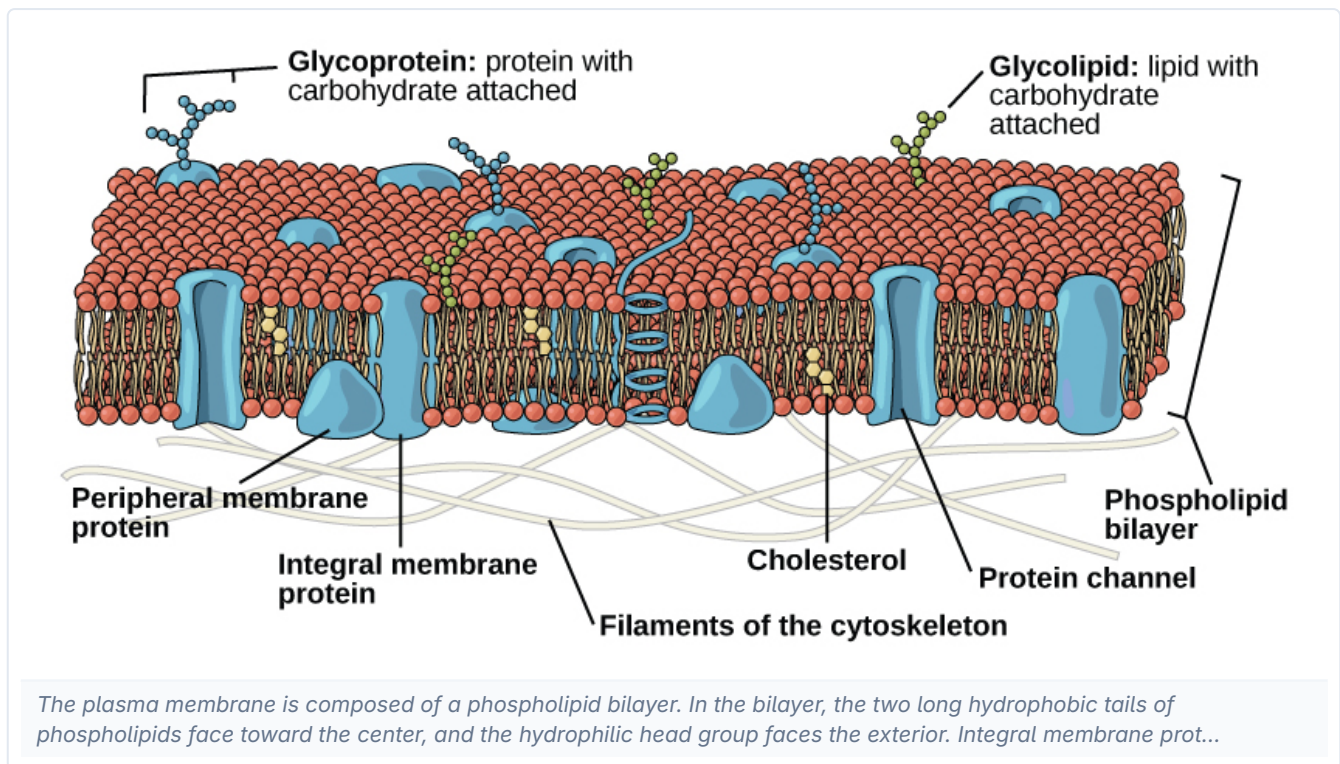


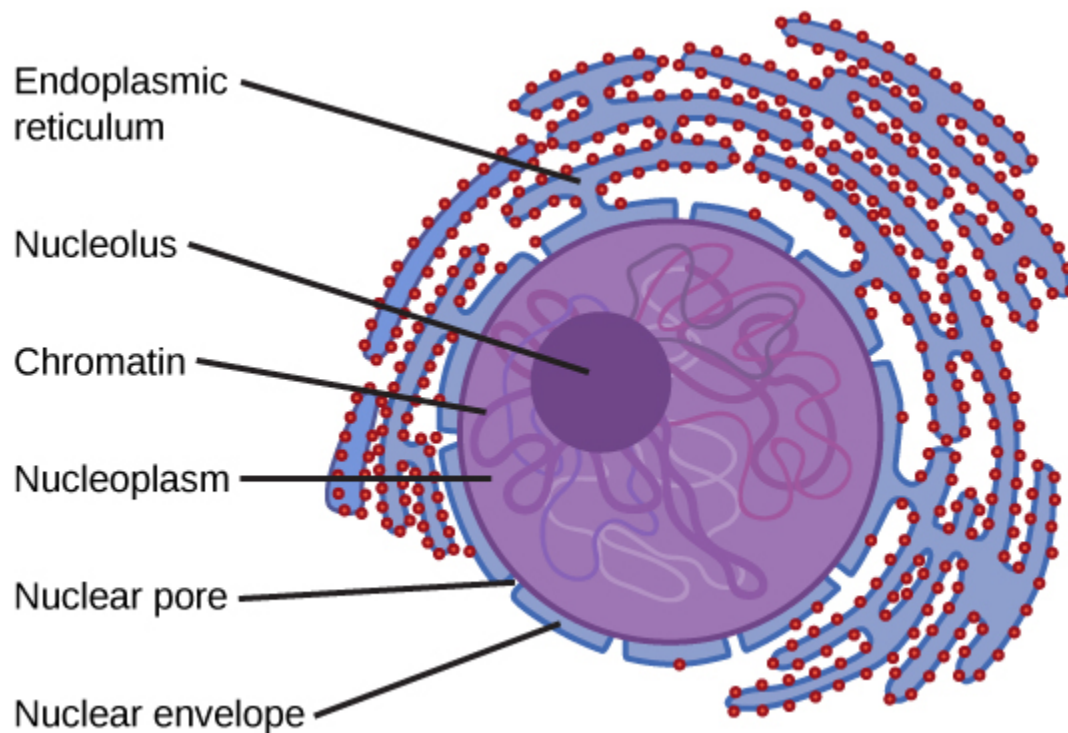
Figure 4.9 The eukaryotic plasma membrane is a phospholipid bilayer with proteins and cholesterol embedded in it.

The Cytoplasm

The cytoplasm is the cell's internal region. It is comprised of organelles suspended in the gel-like cytosol, the cytoskeleton, and various chemicals ([Figure 4.8](#)). Even though the cytoplasm consists of 70 to 80 percent water, it has a semi-solid consistency, which comes from the proteins within it. Many metabolic reactions, including protein synthesis, take place in the cytoplasm.

The Nucleus

Typically, the nucleus is the most prominent organelle in a cell ([Figure 4.8](#)). The nucleus (plural = nuclei) houses the cell's DNA and directs the synthesis of ribosomes and proteins. Let's look at it in more detail ([Figure 4.11](#)).



The two-dimensional image depicts the nucleus of a cell as a circular object with two membranes; several gaps appear in the circle, representing nuclear pores. Surrounding the nucleus are membranous sacks representing th...

Figure 4.11 The nucleus stores chromatin (DNA plus proteins) in a gel-like substance called the nucleoplasm. The nucleolus is a condensed chromatin region where ribosome synthesis occurs. We call the nucleus' boundary the nuclear envelope. It consists of two phospholipid bilayers: an outer and an inner membrane. The nuclear membrane is continuous with the endoplasmic reticulum. Nuclear pores allow substances to enter and exit the nucleus.

The Nuclear Envelope

The nuclear envelope is a double-membrane structure that constitutes the nucleus' outermost portion ([Figure 4.11](#)). Both the nuclear envelope's inner and outer membranes are phospholipid bilayers.

The nuclear envelope is punctuated with pores that control the passage of ions, molecules, and RNA between the nucleoplasm and cytoplasm.

Chromosomes

Chromosomes, structures within the nucleus that are made up of DNA, are located in the nucleus. In eukaryotes, chromosomes are linear structures. Every eukaryotic species has a specific number of chromosomes in the nucleus of each cell. For example, in humans, the chromosome number is 46, while in fruit flies, it is eight. Chromosomes are only visible and distinguishable from one another when the cell is getting ready to divide.

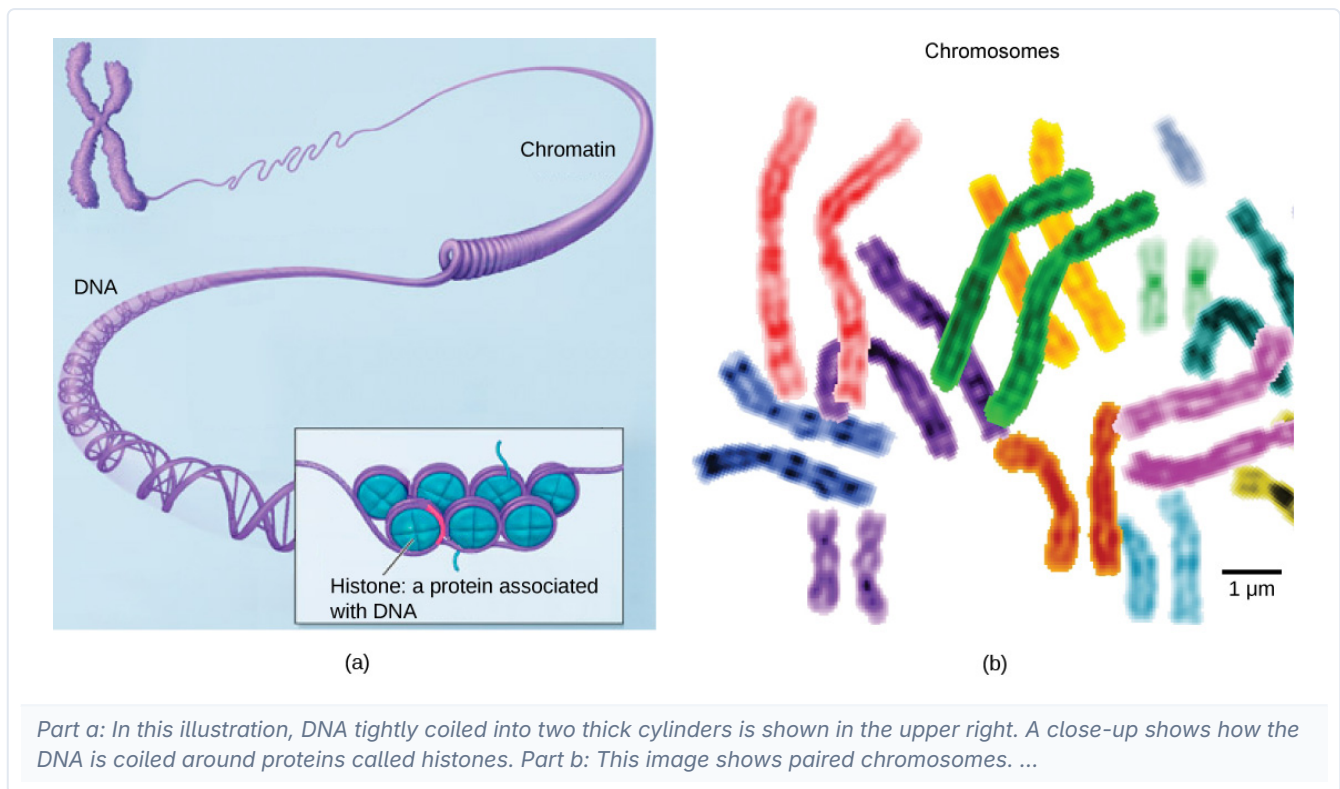


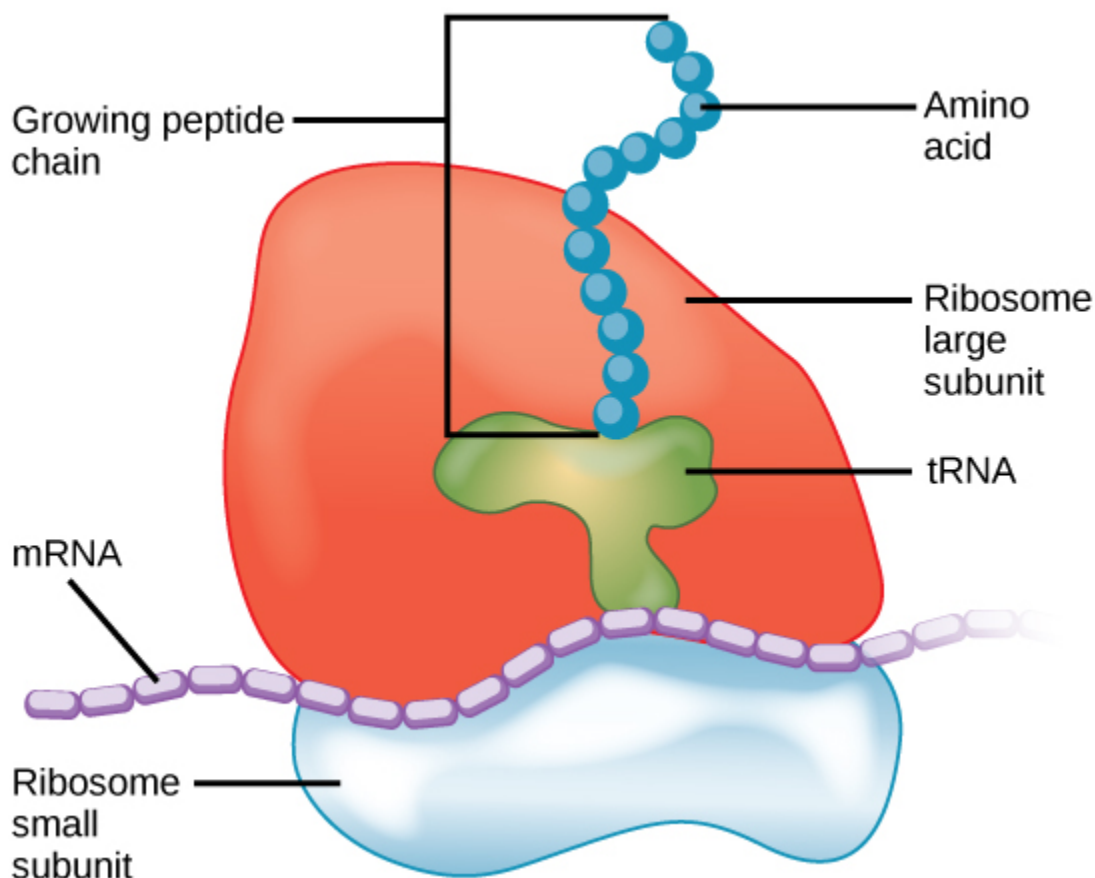
Figure 4.12 (a) This image shows various levels of chromatin's organization (DNA and protein). (b) This image shows paired chromosomes. (credit b: modification of work by NIH; scale-bar data from Matt Russell)

The Nucleolus

We already know that the nucleus directs the synthesis of ribosomes, but how does it do this? The darkly stained area within the nucleus called the nucleolus (plural = nucleoli) aggregates ribosomal RNA with associated proteins to assemble the ribosomal subunits that are then transported out through the pores in the nuclear envelope to the cytoplasm.

Ribosomes

Ribosomes are the cellular structures responsible for protein synthesis. When we view them through an electron microscope, ribosomes appear either as clusters (polyribosomes) or single, tiny dots that float freely in the cytoplasm. They may be attached to the plasma membrane's cytoplasmic side or the endoplasmic reticulum's cytoplasmic side and the nuclear envelope's outer membrane ([Figure 4.8](#)). Electron microscopy shows us that ribosomes consist of two subunits, large and small ([Figure 4.13](#)). Ribosomes receive their "orders" for protein synthesis from the nucleus where the DNA transcribes into messenger RNA (mRNA). The mRNA travels to the ribosomes, which translate the code provided by the sequence of the nitrogenous bases in the mRNA into a specific order of amino acids in a protein. Amino acids are the building blocks of proteins.

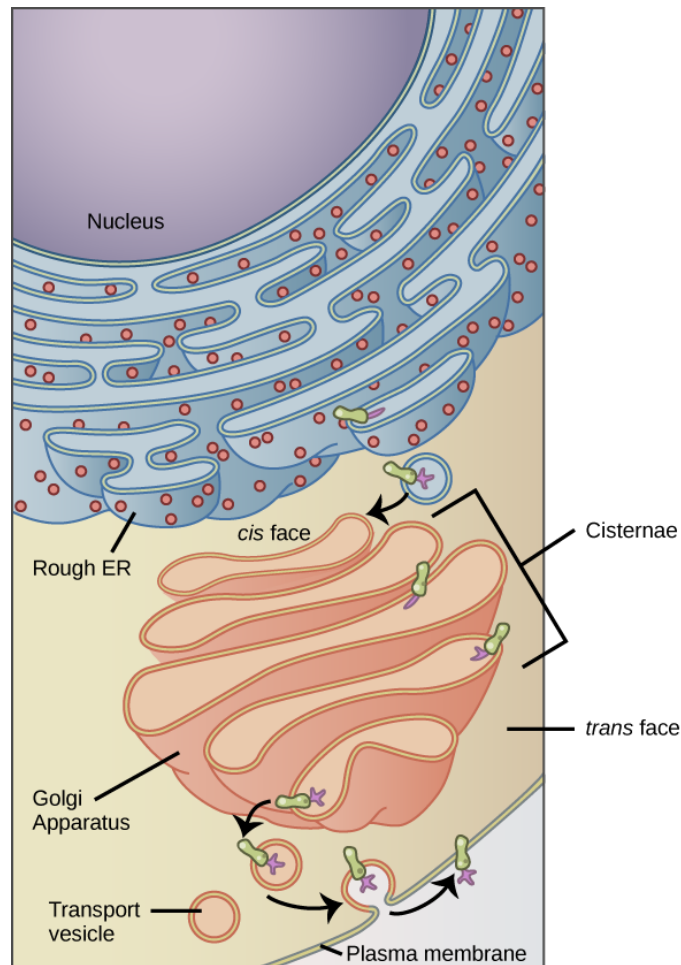


The ribosome consists of a small subunit and a large subunit, which is about three times as big as the small one. The large subunit sits on top of the small one. A chain of m R N A threads between the large and small sub...

Figure 4.13 A large subunit (top) and a small subunit (bottom) comprise ribosomes. During protein synthesis, ribosomes assemble amino acids into proteins.

The Endomembrane System

The endomembrane system (endo = "within") is a group of membranes and organelles ([Figure 4.18](#)) in eukaryotic cells that work together to modify, package, and transport lipids and proteins. It includes the nuclear envelope, the endoplasmic reticulum, the Golgi apparatus, and vesicles. Although not technically *within* the cell, the plasma membrane is included in the endomembrane system because, as you will see, it interacts with the other endomembranous organelles. The endomembrane system does not include either mitochondria or chloroplast membranes.



The left part of this figure shows the rough E R with an integral membrane protein embedded in it. The part of the protein facing the inside of the E R has a carbohydrate attached to it. The protein is shown leaving the ...

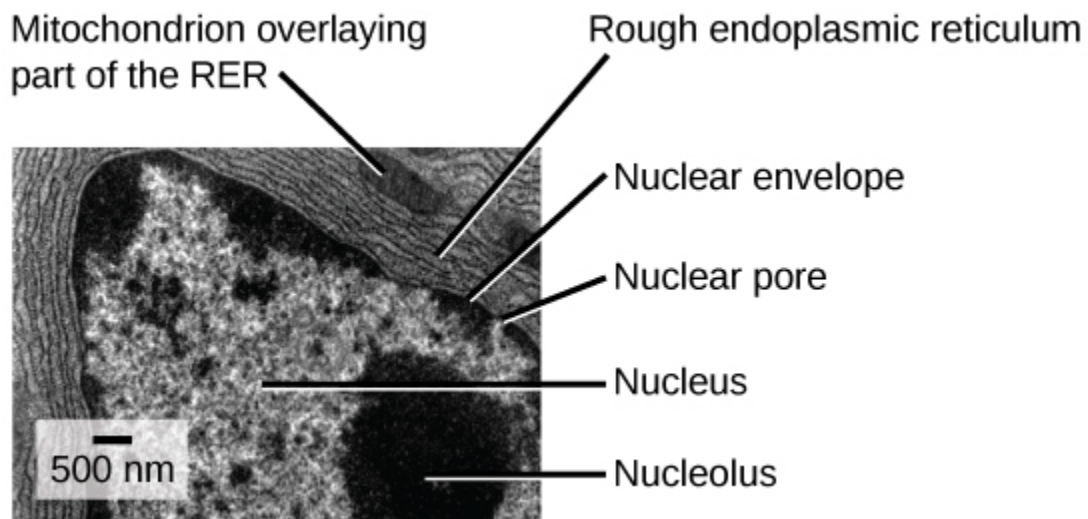
Figure 4.18 Membrane and secretory proteins are synthesized in the rough endoplasmic reticulum (RER). The RER also sometimes modifies proteins. In this illustration, a (green) integral membrane protein is modified by attachment of a (purple) carbohydrate in the ER. Vesicles with the integral protein bud from the ER and fuse with the Golgi apparatus' cis face. As the protein passes along the Golgi's cisternae, the addition of more carbohydrates further modifies it. After its synthesis is complete, it exits as an integral membrane protein of the vesicle that buds from the Golgi's trans face. When the vesicle fuses with the cell membrane, the protein becomes an integral portion of that cell membrane. (credit: modification of work by Magnus Manske)

The Endoplasmic Reticulum

The endoplasmic reticulum (ER) ([Figure 4.18](#)) is a series of interconnected membranous sacs and tubules that collectively modifies proteins and synthesizes lipids. However, these two functions take place in separate areas of the ER: the rough ER and the smooth ER, respectively.

Rough ER

Scientists have named the rough endoplasmic reticulum (RER) as such because the ribosomes attached to its cytoplasmic surface give it a studded appearance when viewing it through an electron microscope ([Figure 4.19](#)).



In this transmission electron micrograph, the nucleus is the most prominent feature. The nucleolus is a circular, dark region inside the nucleus. A nuclear pore can be seen in the nuclear envelope that surrounds the nucl...

Figure 4.19 This transmission electron micrograph shows the rough endoplasmic reticulum and other organelles in a pancreatic cell. (credit: modification of work by Louisa Howard)

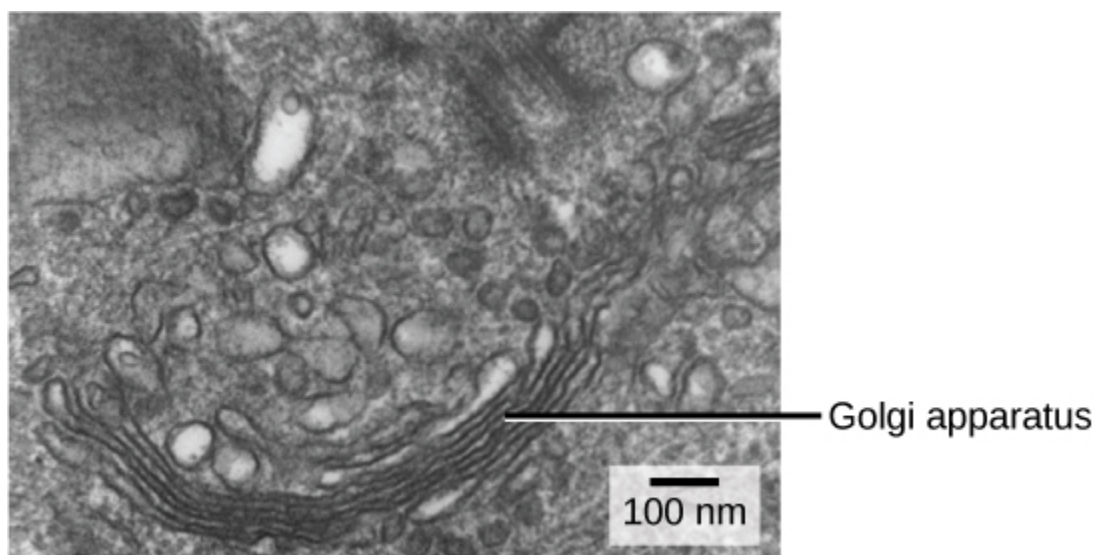
Ribosomes transfer their newly synthesized proteins into the RER's lumen where they undergo structural modifications, such as folding or acquiring side chains. Proteins modified by the RER will be packaged into vesicles that will continue on to the Golgi Apparatus.

Smooth ER

The smooth endoplasmic reticulum (SER) has no ribosomes on its cytoplasmic surface ([Figure 4.18](#)). SER functions include synthesis of lipids and steroid hormones; detoxification of medications and poisons; and storing calcium ions. Lipids produced by the SER will be packaged into vesicles that will continue on to the Golgi Apparatus

The Golgi Apparatus

We have already mentioned that vesicles can bud from the ER and transport their contents elsewhere, but where do the vesicles go? Before reaching their final destination, the lipids or proteins within the transport vesicles still need sorting, packaging, and tagging so that they end up in the right place. Sorting, tagging, packaging, and distributing lipids and proteins takes place in the Golgi apparatus (also called the Golgi body), a series of flattened membranous sacs ([Figure 4.20](#)).



In this transmission electron micrograph, the Golgi apparatus appears as a stack of membranes surrounded by unnamed organelles.

Figure 4.20 The Golgi apparatus in this white blood cell is visible as a stack of semicircular, flattened rings in the lower portion of the image. You can see several vesicles near the Golgi apparatus. (credit: modification of work by Louisa Howard)

The side of the Golgi apparatus that is closer to the ER is called the **cis** face. The opposite side is the **trans** face. The transport vesicles that formed from the ER travel to the **cis** face, fuse with it, and empty their contents into the Golgi apparatus' lumen. As the proteins and lipids travel through the Golgi, they undergo further modifications that allow them to be sorted. The most frequent modification is adding short sugar molecule chains. These newly modified proteins and lipids then tag with phosphate groups or other small molecules in order to travel to their proper destinations.

Finally, the modified and tagged proteins are packaged into secretory vesicles that bud from the Golgi's **trans** face. While some of these vesicles deposit their contents into other cell parts where they will be used, other secretory vesicles fuse with the plasma membrane and release their contents outside the cell.

Lysosomes

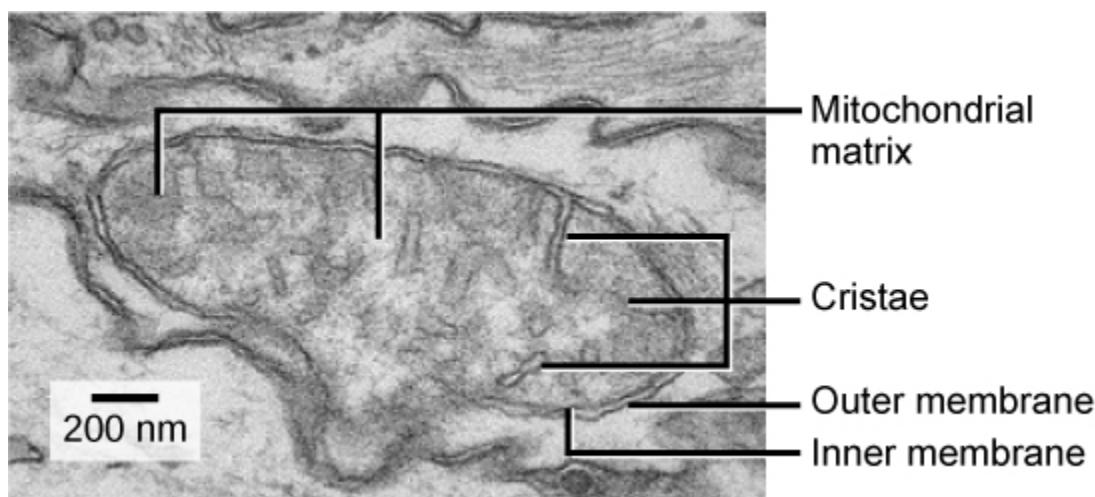
The lysosomes are the cell's "garbage disposal." Enzymes within the lysosomes aid in breaking down proteins, polysaccharides, lipids, nucleic acids, and even worn-out organelles. These enzymes are active at a much lower pH than the cytoplasm's. Therefore, the pH within lysosomes is more acidic than the cytoplasm's pH. Many reactions that take place in the cytoplasm could not occur at a low pH, so again, the advantage of compartmentalizing the eukaryotic cell into organelles is apparent.

Mitochondria

Scientists often call mitochondria (singular = mitochondrion) "powerhouses" or "energy factories" of both plant and animal cells because they are responsible for making adenosine triphosphate (ATP), the cell's main energy-carrying molecule. ATP represents the cell's short-term stored energy. Cellular respiration is the process of making ATP using the chemical energy in glucose and other nutrients. In mitochondria, this process uses oxygen and produces carbon dioxide as a waste product. In fact, the carbon dioxide that you exhale with every breath comes from the cellular reactions that produce carbon dioxide as a byproduct.

In keeping with our theme of form following function, it is important to point out that muscle cells have a very high concentration of mitochondria that produce ATP. Your muscle cells need considerable energy to keep your body moving. When your cells don't get enough oxygen, they do not make much ATP. Instead, producing lactic acid accompanies the small amount of ATP they make in the absence of oxygen.

Mitochondria are oval-shaped, double membrane organelles ([Figure 4.14](#)) that have their own ribosomes and DNA. Each membrane is a phospholipid bilayer. The inner layer has folds called cristae. We call the area surrounded by the folds the mitochondrial matrix. The cristae and the matrix have different roles in cellular respiration.



This transmission electron micrograph of a mitochondrion shows an oval outer membrane and an inner membrane with many folds called cristae. Inside the inner membrane is a space called the mitochondrial matrix.

Figure 4.14 This electron micrograph shows a mitochondrion through an electron microscope. This organelle has an outer membrane and an inner membrane. The inner membrane contains folds, called cristae, which increase its surface area. We call the space between the two membranes the intermembrane space, and the space inside the inner membrane the mitochondrial matrix. ATP synthesis takes place on the inner membrane. (credit: modification of work by Matthew Britton; scale-bar data from Matt Russell)

Peroxisomes

Peroxisomes are small, round organelles enclosed by single membranes. They carry out oxidation reactions that break down fatty acids and amino acids. They also detoxify many poisons that may enter the body. Many of these oxidation reactions release hydrogen peroxide, H_2O_2 , which would be damaging to cells; however, when these reactions are confined to peroxisomes, enzymes safely break down the H_2O_2 into oxygen and water. For example, peroxisomes in liver cells detoxify alcohol.

Vesicles and Vacuoles

Vesicles and vacuoles are membrane-bound sacs that function in storage and transport. Other than the fact that vacuoles are somewhat larger than vesicles, there is a very subtle distinction between them. Vesicle membranes can fuse with either the plasma membrane or other membrane systems within the cell.

Chloroplasts

Like the mitochondria, chloroplasts have their own DNA and ribosomes, but chloroplasts have an entirely different function. Chloroplasts are plant cell organelles that carry out photosynthesis. Photosynthesis is the series of reactions that use carbon dioxide, water, and light energy to make glucose and oxygen. This is a major difference between plants and animals. Plants (autotrophs) are able to make their own food, like sugars used in cellular respiration to provide ATP energy generated in the plant mitochondria. Animals (heterotrophs) must ingest their food. The chloroplasts contain a green pigment, chlorophyll, which captures the light energy that drives the reactions of photosynthesis. Like plant cells, photosynthetic protists also have chloroplasts. Some bacteria perform photosynthesis, but their chlorophyll is not relegated to an organelle.

Like mitochondria, chloroplasts have outer and inner membranes, but within the space enclosed by a chloroplast's inner membrane is a set of interconnected and stacked fluid-filled membrane sacs we call thylakoids ([Figure 4.17](#)). Each thylakoid stack is a granum (plural = grana). We call the fluid enclosed by the inner membrane that surrounds the grana the stroma.

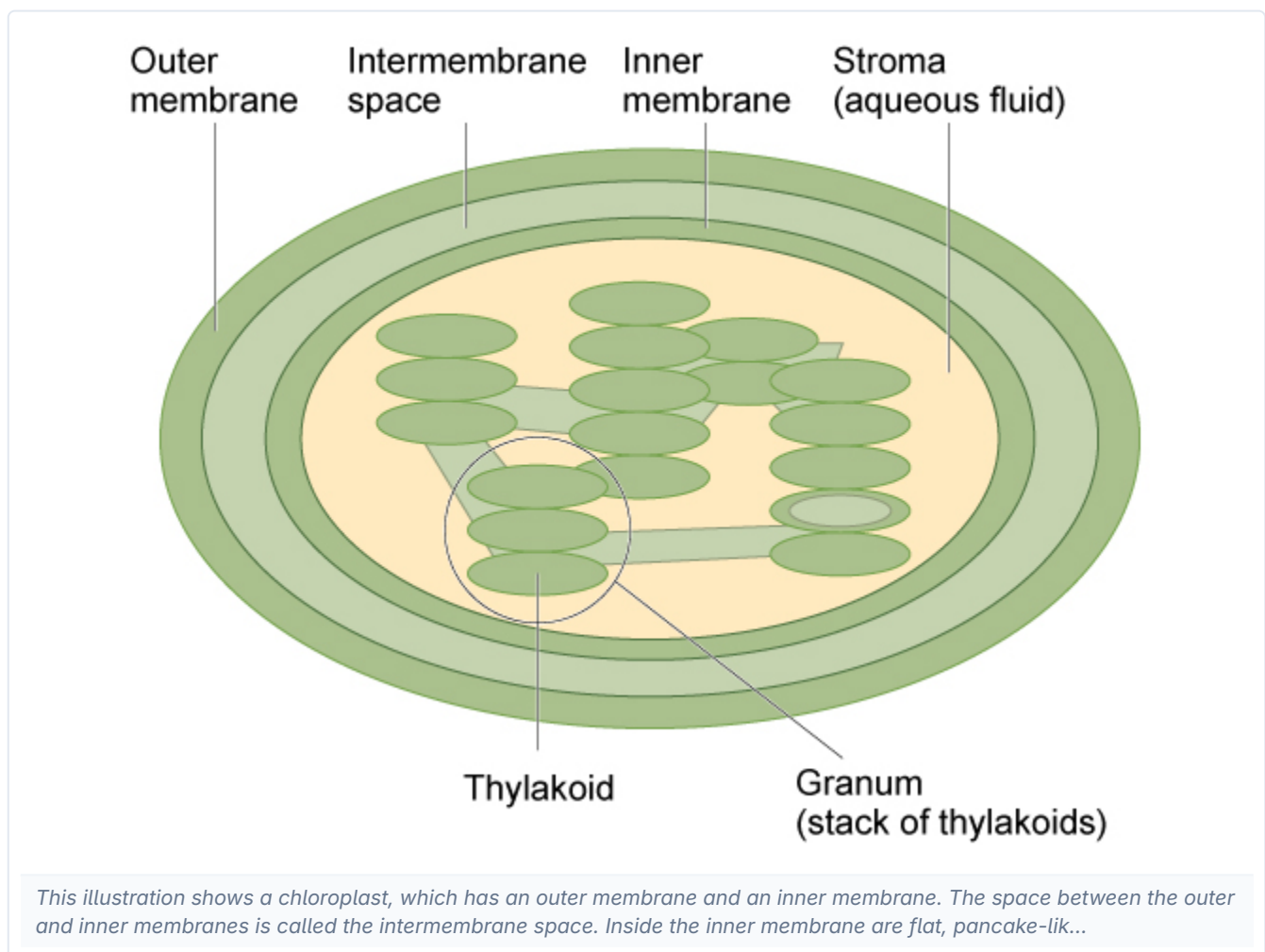


Figure 4.17 The chloroplast has an outer membrane, an inner membrane, and membrane structures - thylakoids that are stacked into grana. We call the space inside the thylakoid membranes the thylakoid space. The light harvesting reactions take place in the thylakoid membranes, and sugar synthesis takes place in the fluid inside the inner membrane, which we call the stroma. Chloroplasts also have their own genome, which is contained on a single circular chromosome.

EVOLUTION CONNECTION

Endosymbiosis

We have mentioned that both mitochondria and chloroplasts contain DNA and ribosomes. Have you wondered why? Strong evidence points to endosymbiosis as the explanation.

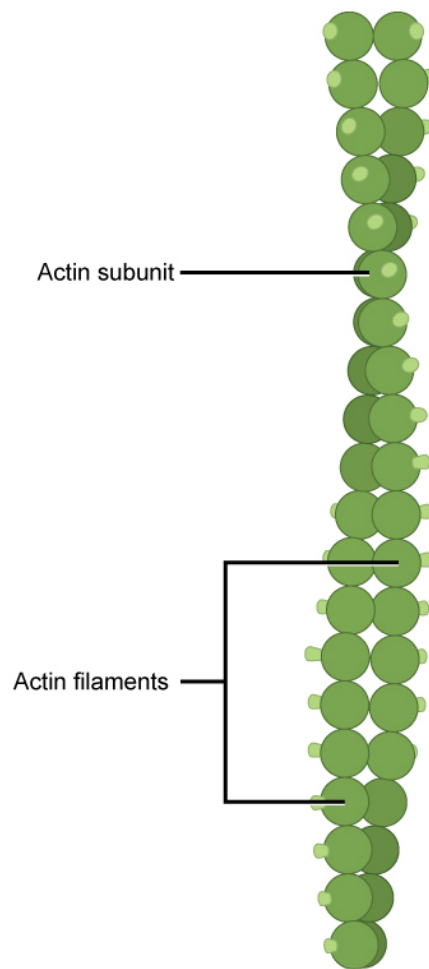
Symbiosis is a relationship in which organisms from two separate species depend on each other for their survival. Endosymbiosis (endo- = “within”) is a mutually beneficial relationship in which one organism lives inside the other. Endosymbiotic relationships abound in nature. We have already mentioned that microbes that produce vitamin K live inside the human gut. This relationship is beneficial for us because we are unable to synthesize vitamin K. It is also beneficial for the microbes because they are protected from other organisms and from drying out, and they receive abundant food from the environment of the large intestine.

Scientists have long noticed that bacteria, mitochondria, and chloroplasts are similar in size. We also know that bacteria have DNA and ribosomes, just like mitochondria and chloroplasts. Scientists believe that host cells and bacteria formed an endosymbiotic relationship when the host cells ingested both aerobic and autotrophic bacteria (cyanobacteria) but did not destroy them. Through many millions of years of evolution, these ingested bacteria became more specialized in their functions, with the aerobic bacteria becoming mitochondria and the autotrophic bacteria becoming chloroplasts.

The Cytoskeleton

Microfilaments

Of the three types of protein fibers in the cytoskeleton, microfilaments are the narrowest. They function in cellular movement, have a diameter of about 7 nm, and are comprised of two globular protein intertwined strands, which we call actin ([Figure 4.23](#)). For this reason, we also call microfilaments actin filaments.



This illustration shows two actin filaments wound together. Each actin filament is composed of many actin subunits connected together to form a chain.

Figure 4.23 Two intertwined actin strands comprise microfilaments.

ATP powers actin to assemble its filamentous form, which serves as a track for the movement of a motor protein we call myosin. This enables actin to engage in cellular events requiring motion, such as cell division in eukaryotic cells and cytoplasmic streaming, which is the cell cytoplasm's circular movement in plant cells. Actin and myosin are plentiful in muscle cells. When your actin and myosin filaments slide past each other, your muscles contract.

Microfilaments also provide some rigidity and shape to the cell. They can depolymerize (disassemble) and reform quickly, thus enabling a cell to change its shape and move. White blood cells (your body's infection-fighting cells) make good use of this ability. They can move to an infection site and phagocytize the pathogen.

Intermediate Filaments

Several strands of fibrous proteins that are wound together comprise intermediate filaments ([Figure 4.24](#)). Intermediate filaments get their name from the fact that their diameter, 8 to 10 nm, is between those of microfilaments and microtubules.



This illustration shows 10 intermediate filament fibers bundled together.

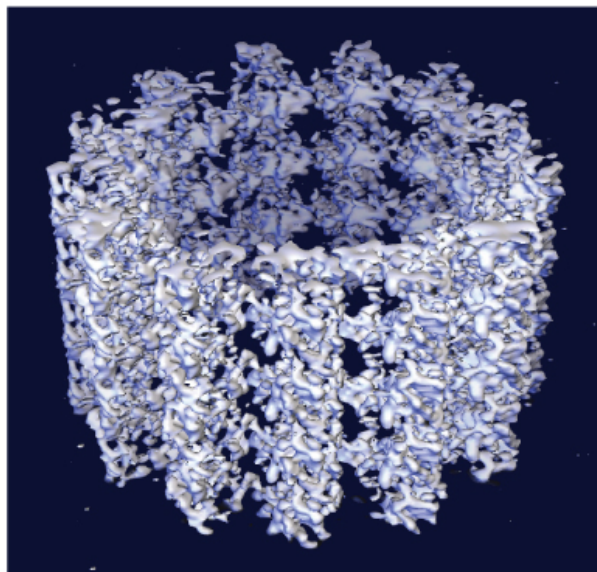
Figure 4.24 Intermediate filaments consist of several intertwined strands of fibrous proteins.

Intermediate filaments have no role in cell movement. Their function is purely structural. They bear tension, thus maintaining the cell's shape, and anchor the nucleus and other organelles in place. [Figure 4.22](#) shows how intermediate filaments create a supportive scaffolding inside the cell.

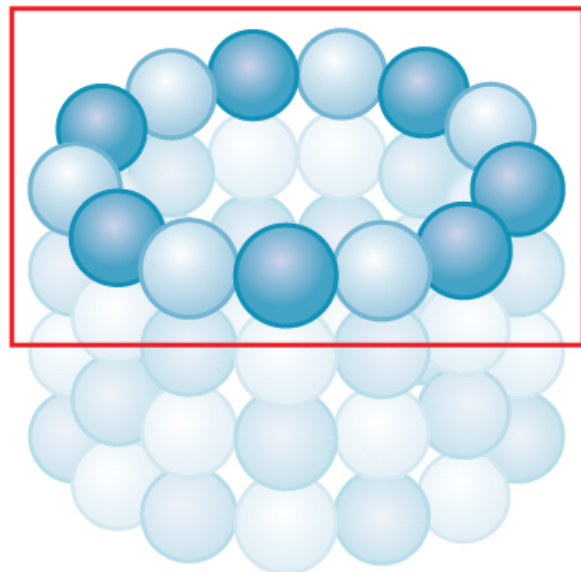
The intermediate filaments are the most diverse group of cytoskeletal elements. Several fibrous protein types are in the intermediate filaments. You are probably most familiar with keratin, the fibrous protein that strengthens your hair, nails, and the skin's epidermis.

Microtubules

As their name implies, microtubules are small hollow tubes. Polymerized dimers of α -tubulin and β -tubulin, two globular proteins, comprise the microtubule's walls ([Figure 4.25](#)). With a diameter of about 25 nm, microtubules are cytoskeletons' widest components. They help the cell resist compression, provide a track along which vesicles move through the cell, and pull replicated chromosomes to opposite ends of a dividing cell. Like microfilaments, microtubules can disassemble and reform quickly.



13 polymerized dimers
of α -tubulin and β -tubulin



The left part of this figure is a molecular model of 13 polymerized dimers of alpha- and beta-tubulin joined together to form a hollow tube. The right part of this image shows the tubulin structure as a ring of spheres c...

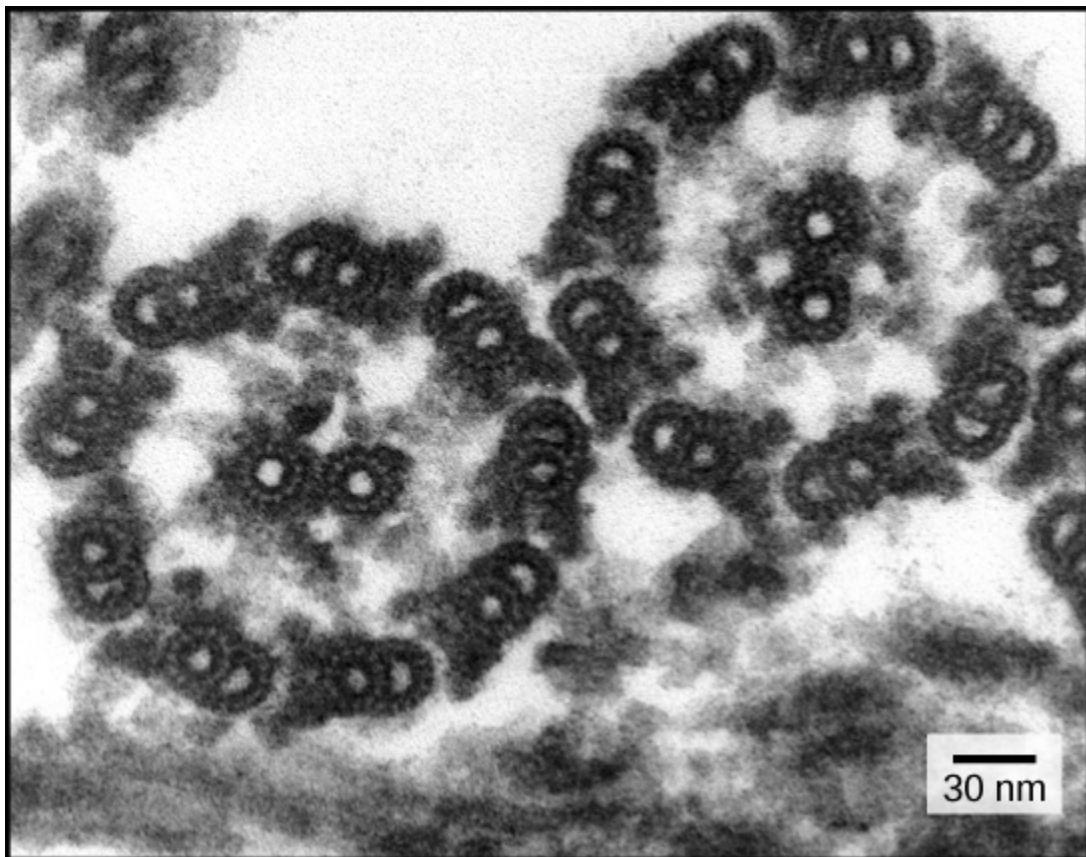
Figure 4.25 Microtubules are hollow. Their walls consist of 13 polymerized dimers of α -tubulin and β -tubulin (right image). The left image shows the tube's molecular structure.

Microtubules are also the structural elements of flagella, cilia, and centrioles (the latter are the centrosome's two perpendicular bodies). In animal cells, the centrosome is the microtubule-organizing center. In eukaryotic cells, flagella and cilia are quite different structurally from their counterparts in prokaryotes, as we discuss below.

Flagella and Cilia

The flagella (singular = flagellum) are long, hair-like structures that extend from the plasma membrane and enable an entire cell to move (for example, sperm, *Euglena*, and some prokaryotes). When present, the cell has just one flagellum or a few flagella. However, when cilia (singular = cilium) are present, many of them extend along the plasma membrane's entire surface. They are short, hair-like structures that move entire cells (such as paramecia) or substances along the cell's outer surface (for example, the cilia of cells lining the Fallopian tubes that move the ovum toward the uterus, or cilia lining the cells of the respiratory tract that trap particulate matter and move it toward your nostrils.)

Despite their differences in length and number, flagella and cilia share a common structural arrangement of microtubules called a "9 + 2 array." This is an appropriate name because a single flagellum or cilium is made of a ring of nine microtubule doublets, surrounding a single microtubule doublet in the center ([Figure 4.26](#)).



This transmission electron micrograph shows a cross section of nine microtubule doublets that form a hollow tube. Another microtubule doublet sits in the center of the tube.

Figure 4.26 This transmission electron micrograph of two flagella shows the microtubules' 9 + 2 array: nine microtubule doublets surround a single microtubule doublet. (credit: modification of work by Dartmouth Electron Microscope Facility, Dartmouth College; scale-bar data from Matt Russell)